

PERSPECTIVE

Programming of CAR-macrophages for cancer therapy

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Intra Tumoral Injection of Nanocomplex-Mediated In Vivo Programming Agents to Mediate Chimeric Antigen Receptor-M1 Macrophages for Cancer Therapy

Adoptive cell immunotherapy (ACT) is a promising field for the treatment of cancer. A prominent example of this concept is the use of chimeric antigen receptor (CAR)-T cell therapy, which was shown in the past years to be an efficient treatment for hematological malignancies.[1,2] CAR is an engineered receptor that enhances recognition of cancer cell surface antigen and induces downstream signalling of immune cells to attack the tumor.[3] Despite proof of efficacy, CAR T-cell therapy requires the manufacturing of autologous CAR-T cells ex vivo and thus remains costly.[4] Moreover, CAR T-cell therapy is inefficient on solid tumors. Reasons for this are that T-cell infiltration rate in solid tumors is low and that the immunosuppressive tumor microenvironment (TME) of solid tumors inactivates T cells.[5]

In this issue, Zhang et al. have developed induced pluripotent stem cells (iPSCs)-derived, CAR-expressing macrophage cells (CAR-iMac). The expression of CAR gives antigen-dependent functions to macrophages including polarization towards pro-inflammatory state, expression and secretion of cytokines, enhanced phagocytosis and overall in vivo anti-cancer cell activity.

Macrophage is the innate immune cell with the highest infiltration rate in the tumor microenvironment. Therefore, it has recently emerged as an interesting candidate for solid tumors therapy. Mononuclear-phagocytes system (MPS), consisting of monocytes and monocytes-derived macrophages, selectively

engulf external targets upon “non-self” antibody recognition.[6] However, genetically unstable tumor cells use self-labelling as a general mechanism to avoid phagocytosis.[7]

Within tumor tissues, M1 macrophages (pro-inflammatory) polarize into M2-type (immunosuppressive) and prevent immune response.[8] In addition, cancer cells secrete cytokines (interleukin (IL)-6 and transforming growth factor (TGF)- β). These cytokines, associated with the hypoxic environment of tumor tissues, influence the tumor-associated macrophages (TAM) to up regulate angiogenesis, thus promoting cancer growth.[9,10] These M2 TAM constitute a large portion of the tumor and play a critical role in cancer development.

Historical attempts to fight cancer with macrophages showed little effects in clinical trials. This suggests that macrophages require additional signals to target tumors.[11] Hopefully, CAR-M opens up new possibilities for immunotherapy. By transferring specifically engineered CAR genes into macrophages, researchers can equip them with the enhanced ability to bind the cancer cell surface via specific antigen identification and subsequently activate anti-tumor activity.[12]

Immortalized macrophage cell lines are not suitable for clinical settings, and peripheral blood mononuclear cells (PBMC) or bone marrow-derived macrophages are difficult to efficiently engineer. To address these challenges, Zhang et al. created CAR-M cells from pluripotent stem cells (iPSC) derived

from PBMC of a healthy human donor. CD19-CARs were selected and introduced into the iPSCs by lentiviral transduction. After induced differentiation toward myeloid cell lineage, mature cells displayed key macrophage marker genes (AIF1, CSF1R and SPI1) while pluripotent makers were depleted (LIN28A and POU5F1). Single-cell RNA sequencing analysis was performed to separate CAR-iMacs from undifferentiated CAR-iPSCs. CAR-iMac cell populations appeared relatively homogeneous and blasting against human cell atlas revealed high clustering with macrophage with convergence towards M2 state.

To test if CAR-iMac could target a specific antibody and “switch” to pro-inflammatory state upon recognition, CAR(CD19)-iMacs and CAR(meso)-iMacs were incubated respectively with CD19-expressing K562 leukemia cells and mesothelin-expressing OVCAR3/ASPC1 ovarian/pancreatic cancer cells. Results highlighted that K562-CD19 cells were more likely to be phagocytosed by CAR(CD19)-iMacs than K562 alone, and that CAR(meso)-iMac showed increased phagocytosis activity against OVCAR3 and ASPC1 cells compared to control-iMacs. Cytokine gene expression and transcriptional analysis revealed antigen-dependent increase in M1 pro-inflammatory cytokine expression and strong enrichment of up-regulated genes. This seems to indicate that CAR-iMacs can polarize to pro-inflammatory state when they encounter a specific antigen.

To study anti-tumor activity, CAR-iMacs were administered to NSG mice previously injected with ovarian cancer cells (HO8910). These cells expanded in-vivo for 3 days (2-fold) and disappeared after around 30 days. Before injection, CAR-iMacs were treated with IFN- γ to further polarize the cells toward M1-type. Interestingly, CAR(meso)-iMac-treated mice showed reduced tumor burden compared to the control group.

Overall, iPSC-derived macrophages may become an important source of cells for myeloid-based cancer immunotherapy. However, this approach still requires the fastidious process of ex vivo CAR-M production, which limits its broader clinical applications.

To overcome both the costly process of ex-vivo production and the tumorigenicity concerns associated with viral transduction, Kang et al. reported an in vivo repro-

gramming procedure of tumor-associated M2-macrophage expressing CAR and anti-tumor M1 phenotype. Genes encoding CAR and IFN- γ (used to further promote the pro-inflammatory M1 phenotype) were delivered to native macrophages using nanocarriers. Mannose receptors are over-expressed in macrophages.[13] Therefore, mannose-conjugated polyethylenimine (MPEI) was used as the gene delivery carrier for macrophage-targeting. ALK antigen was chosen for neuroblastoma tumor-specific targeting (ALK is overexpressed on the surface of neuroblastoma). To build CAR-IFN- γ gene constructs, anti-ALK and CAR-IFN- γ genes were cloned into non-viral piggyBac transposon vectors. Nanocomplexes of MPEI and CAR-IFN- γ pDNA (MPEI/pCAR-IFN- γ) were then formed via electrostatic interactions. Stability analysis showed that MPEI/pCAR-IFN- γ was stable for 7 days in serum and transfection efficiency was evaluated at ~14%, which is typically lower than lentiviral transduction.

MPEI/pCAR-IFN- γ was intratumorally injected into Neuro-2a tumor-bearing A/J mice. After 48h, the stability of CAR gene expression in the tumors was validated by in-vivo fluorescence imaging. Following the second injection (72h after the first one), CAR expression was not detectable in the major organs. Serum levels of enzymes reflecting the functions of the liver and kidney were evaluated and showed no systemic toxicity for the therapy. Flow cytometry and immunohistochemistry (IHC) showed that most of the CAR expression occurred in macrophages, followed by dendritic cells. This seems to show that MPEI can efficiently target delivery of pCAR-IFN- γ to macrophages. The difference of CAR expression between macrophages and dendritic cells (both of which express mannose receptor) may be because macrophages are predominant in tumors compared to dendritic cells.[14]

Mice injected intratumorally with MPEI/pCAR-IFN- γ displayed delayed tumor growth compared to control groups (MPEI/pCD19, MPEI/pCAR or MPEI/pIFN- γ). Interestingly, 50% of mice in the MPEI/pCAR-IFN- γ group survived more than 100 days after tumor modelling. The average body weight of mice did not significantly change between groups, suggesting minimal side effects associated with the treatments. H&E staining coupled with TUNEL assay

revealed that they were fewer cancer cells and more apoptotic cells in the MPEI/pCAR-IFN- γ -treated group. This suggests CAR-mediated phagocytosis of cancer cells and CAR-M1 macrophage-mediated enhanced activation of cytotoxic T cells (CD8 $^{+}$ T cells). Altogether, these results seem to indicate that CAR and IFN- γ genes additively exert anti-tumor effects.

To better understand the immune mechanisms underlying the anti-tumor effects of MPEI/pCAR-IFN- γ , Kang et al. performed flow cytometry to analysed immune cells in the tumor. Results showed that ~10% of intra-tumoral macrophages expressed CAR. IHC and flow cytometry confirmed an increased expression of M1 macrophage marker (CD86 and iNOS) and a decrease in M2 macrophage markers (Arginase-1 and CD163). The high expression of pro-inflammatory inducible nitric oxide synthase (iNOS) could be explained by the fact that it is directly induced by IFN- γ .^[15] Intriguingly, the number of macrophages increased in all mice injected with MPEI. An explanation for this may be that MPEI exerts an adjuvant effect when injected.^[16] Moreover, the injection of MPEI/pCAR-IFN- γ suppressed the expression of anti-inflammatory cytokines (IL-4 and IL-10) in the tumor. This phenomenon usually correlates with a decreasing number of M2 macrophages and Treg cells in the TME.^[17]

The pro-inflammatory effects of MPEI/pCAR-IFN- γ -induced CAR-M1 macrophages were also shown to affect neighboring immune cells. As both phagocytic and professional anti-gene presenting cells, M1

macrophages can present tumoral phagocytosed antigen to CD8 $^{+}$ T cells and mediate their activation.^[18,19] In mice treated with MPEI-CAR therapy, Kang et al. reported an increase of the number of CD8 $^{+}$ T cells with activation markers (IFN- γ or granzyme b expression). This could be attributed to the CAR-mediated tumor cell phagocytosis and activation of CD8 $^{+}$ T cells by CAR-M1 macrophages.

Overall, Kang et al. demonstrated that intra-tumoral injection of MPEI/pCAR-IFN- γ can induce CAR-M1 macrophages in vivo for effective anti-tumor immunity. This innovative approach may overcome some of the challenges currently associated with CAR-T therapy, such as the low infiltration rate of T-cells in solid tumors or the high cost of CAR immune cell production. The simple process of MPEI/ pCAR-IFN- γ nanocomplex manufacturing might constitute a great opportunity to boost CAR technology for widespread application in cancer therapy.

Although CAR seems to confer anti-cancer activities to macrophages, the efficiency of the therapy could be further improved by designing more effective CAR structures or gene modifications to keep CAR-M closer to the M1 state. Another approach could be to test the association of CAR-T and CAR-M as treatment for solid tumors. Because CAR-T cell therapy is in part limited by T-cell inactivation from the pro-tumoral M2-macrophages, using MPEI/pCAR-IFN- γ to mediate macrophage polarization to M1-phenotype could greatly improve CAR-T anti-tumor effect.

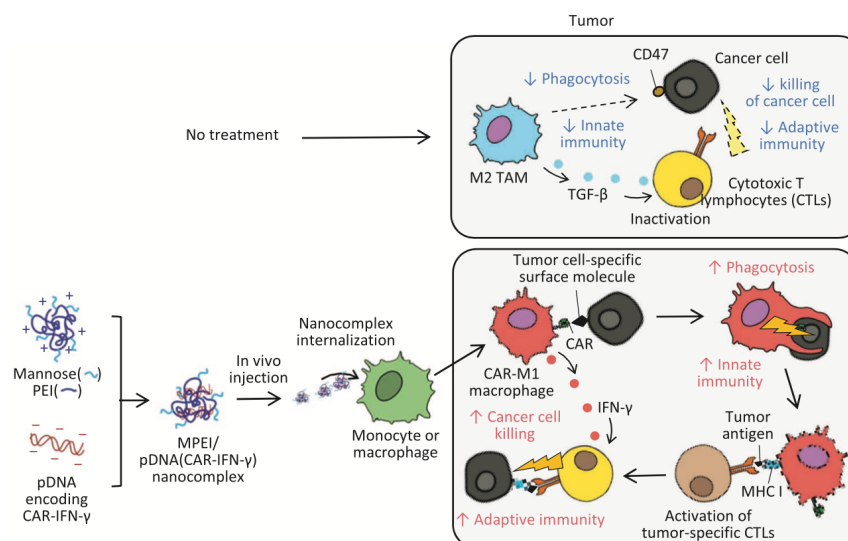


Figure 1. Schematic representation describing the delivery of plasmid DNA encoding ALK-specific CAR and IFN- γ (pCAR-IFN- γ) using MPEI to induce CAR-M1 macrophages in vivo, as well as the anti-tumoral mechanisms of CAR-M1 macrophages expressing CAR and IFN- γ .

Source: Kang et al.

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